Awais Asghar

AI Engineer

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Interests

Machine Learning, Deep Learning, Computer Vision, Generative AI, LLMs, RAG, Data Science, Agentic AI

Education

National University of Sciences and Technology (NUST) Islamabad, Pakistan

Bachelor of Engineering in Electrical Engineering *Sept 2022 - June 2026*

- CGPA: 3.63/4.0 Specialization GPA: 3.94/4.0 Merit Scholarship, 2024 - 2026
- Senior Design Thesis: ML-Based Hardware Accelerator for Real-Time Image Segmentation on FPGA

Research Experience

Mitacs Globalink Research Intern Summer 2026

NRC Herzberg Institute of Astrophysics *Victoria, BC, Canada*

- Conduct applied ML and AI research on large-scale astrophysical datasets, contributing to model development, scientific computing workflows, and data analysis pipelines under Dr. Sebastien Fabbro
- Carrying out end-to-end academic research: literature review, experiment design, result analysis, and scientific documentation

Deep Learning Research Intern June 2024 - Sept 2024

Deep Learning Lab, SINES, NUST *Islamabad, Pakistan*

- Built and evaluated CNN and transfer learning models in PyTorch and TensorFlow with GPU-accelerated training via CUDA, focused on performance benchmarking and model validation

Machine Learning Research Intern June 2023 - Aug 2023

HamsanTech, NSTP *Islamabad, Pakistan*

- Built a multi-model skin cancer classification pipeline achieving 91% accuracy and 0.963 AUC using an ensemble of XGBoost, AdaBoost, LightGBM, SVM, and Logistic Regression
- Developed interactive Power BI and Tableau dashboards and ran Excel-based data cleaning and exploratory analysis on real industry datasets

Selected Projects

ML-Based Hardware Accelerator for Image Segmentation | *Python, PyTorch, Vitis HLS, Vivado* 2025 - 2026

- Achieved 56x model compression to 570 KB with 97% pixel accuracy at 2.72W power on a Zybo Z7-20 FPGA, covering the full pipeline from training and quantization to edge deployment

AI-Based Exam Generator (RAG) | *Python, LangChain, Groq API, Llama 3.1* | Github Repo 2025

- Built an end-to-end RAG pipeline combining document ingestion, vector embeddings, semantic retrieval, and LLM-based question generation into a production-ready system

Lightweight U-Net for Driving Scene Segmentation | *Python, PyTorch, OpenCV* | Github Repo Dec 2025

- Compressed a U-Net from 8.7M to 0.53M parameters while holding 72% IoU, 80% F1, and 96% test accuracy on the CARLA simulator dataset; paper in progress

Real-Time Fabric Defect Detection | *Python, OpenCV, NumPy, Jetson Nano* | Github Repo Dec 2025

- Built a classical computer vision pipeline (GLCM, FFT, Gabor filters, IoU-based fusion) for interpretable, low-power defect localization on edge hardware, with no deep learning required; paper in progress

RecycleVision: Automated Waste Classification | *TensorFlow, Keras, EfficientNetB0* | Github Repo Nov 2025

- Trained an EfficientNetB0 transfer learning model reaching 92.9% test accuracy and a 0.93 macro F1 score across multiple waste categories

SkinSense: Multi-Model Skin Cancer Classifier | *Scikit-learn, XGBoost, PyTorch* | Github Repo 2023

- Built an ensemble classification pipeline achieving 91% accuracy and 0.963 AUC for single-image RGB skin cancer detection

Technical Skills

Machine Learning & AI: Python, TensorFlow, PyTorch, Keras, Scikit-learn, NumPy, Pandas, OpenCV, SciPy, CUDA
LLMs & RAG: LangChain, Groq API, Llama 3.1, Vector Databases, Retrieval-Augmented Generation, Prompt Engineering
Data Analysis & Visualization: Excel, Power BI, Tableau, Matplotlib, Seaborn
Tools & Platforms: Git/GitHub, Linux, Jupyter Notebook, Google Colab, VS Code, Kaggle, LaTeX

Publications (In Progress)

Design of a Novel Lightweight U-Net Architecture for Efficient Semantic Segmentation: based on FYP and autonomous driving segmentation research at NUST SEECS
Real-Time Fabric Defect Detection on Edge Devices Using Classical Computer Vision: multi-method pipeline deployed on Jetson Nano

Certifications and Relevant Coursework

Machine Learning Specialization (3-Course Series), deeplearning.ai: supervised ML, advanced learning algorithms, unsupervised learning, reinforcement learning
Deep Learning Specialization (5-Course Series), deeplearning.ai: neural networks, hyperparameter tuning, CNNs, RNNs, LSTMs, sequence modeling, NLP
Deep Learning Course Series, MIT 6.S191: computer vision, generative modeling, transformers, LLMs, deep reinforcement learning, AI in biology
NUST Coursework: Machine Learning, Deep Learning, Computer Vision, Linear Algebra, Calculus, Digital Signal Processing, Signals and Systems

Honors and Achievements

- Mitacs Globalink Research Internship: fully funded international placement at NRC Herzberg, Canada (Summer 2026)
- Millennium Fellow: selected among the top 4% globally by the UN Academic Impact and Millennium Campus Network
- NUST SEECS Merit Scholarship, 2024 - 2026
- Dean's Honor List: recognized 4 times for outstanding academic performance
- Prime Minister Youth Laptop Scheme, 2025, awarded for academic excellence
- 3rd place, BISE Multan Pre-Engineering, among 70,000+ students
- Intermediate Scholarship: 100% fee waiver based on exceptional academic performance